

SUPERIOR BONG, WI, USA

WMO: 726427

Lat: 46.689N

Lon: 92.094W

Elev: 205

StdP: 98.88

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 54908

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.0	-22.9	-29.9	0.2	-25.0	-27.7	0.3	-22.2	10.3	-8.1	9.0	-8.2	2.7	230	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	30.1	22.2	28.0	20.7	27.1	20.0	23.9	28.3	22.4	26.7	21.1	24.8	4.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.4	17.6	26.8	21.2	16.2	25.1	19.7	14.8	23.6	72.6	28.5	67.1	26.4	62.1	24.1	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.3	7.5	DB	-29.1	33.5	3.1	1.5	-31.3	34.6	-33.1	35.5	-34.9	36.4	-37.1	37.5	(4)
(5)			WB	-29.3	25.7	3.0	1.5	-31.5	26.8	-33.3	27.7	-35.0	28.5	-37.2	29.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.9	-10.1	-8.6	-2.3	3.8	9.3	14.3	19.2	18.3	13.9	7.2	0.3	-7.2	(6)	
	DBStd	11.22	7.13	6.43	6.22	4.03	4.17	4.12	3.41	3.23	4.13	4.18	5.48	6.47	(7)	
	HDD10.0	2736	622	521	381	192	66	8	0	0	11	110	293	533	(8)	
	HDD18.3	5018	881	754	638	437	282	132	32	41	143	347	541	791	(9)	
	CDD10.0	878	0	0	1	5	43	137	285	257	128	22	2	0	(10)	
	CDD18.3	119	0	0	0	0	2	11	58	39	9	1	0	0	(11)	
	CDH23.3	1402	0	0	0	4	63	193	636	387	108	11	0	0	(12)	
	CDH26.7	381	0	0	0	1	15	46	192	103	23	2	0	0	(13)	
(14) Wind	WSAvg	3.1	3.1	3.2	3.1	3.7	3.4	2.9	2.7	2.5	2.7	3.1	3.2	3.0	(14)	
(15) Precipitation	PrecAvg	774	23	23	36	65	78	107	112	93	93	82	46	29	(15)	
	PrecMax	990	53	54	78	222	143	170	254	192	185	177	114	59	(16)	
	PrecMin	535	2	4	11	16	28	35	46	33	25	19	1	4	(17)	
	PrecStd	114	15	13	19	46	29	39	51	42	43	41	33	14	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.1	7.9	16.1	22.4	28.6	31.0	32.7	32.2	29.0	24.6	17.7	6.9	(19)	
		MCWB	3.7	3.8	9.1	12.3	17.8	23.3	23.8	23.6	21.4	16.0	11.2	3.5	(20)	
	2%	DB	2.9	4.4	11.4	17.7	24.6	27.6	30.9	29.1	26.6	19.8	12.9	4.0	(21)	
		MCWB	1.4	1.6	6.7	9.9	15.2	19.5	23.1	22.0	19.7	13.9	8.5	2.6	(22)	
	5%	DB	1.4	2.4	7.7	13.9	21.2	25.9	28.8	27.4	23.8	17.2	10.3	2.6	(23)	
		MCWB	0.5	0.7	4.1	7.5	13.6	18.0	21.5	20.7	18.1	12.4	7.0	1.6	(24)	
	10%	DB	-0.1	0.9	5.3	11.0	17.7	22.8	27.3	25.5	21.5	14.0	7.9	1.5	(25)	
		MCWB	-0.9	-0.4	2.6	6.5	11.7	16.7	20.6	19.6	17.2	10.3	5.8	0.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	4.9	10.9	14.1	19.2	24.2	25.7	25.5	23.4	17.7	12.9	4.3	(27)	
		MCDB	6.5	8.1	15.1	19.7	26.4	29.3	30.2	29.5	27.9	22.1	16.9	5.4	(28)	
	2%	WB	1.9	2.5	6.9	10.8	16.5	21.1	24.2	23.4	21.1	15.1	9.5	2.9	(29)	
		MCDB	2.8	4.2	10.6	17.3	22.0	25.0	28.5	27.7	24.2	18.7	12.2	3.8	(30)	
	5%	WB	0.6	1.0	4.8	8.4	14.5	19.1	22.8	22.0	19.4	12.8	7.7	1.9	(31)	
		MCDB	1.5	2.3	7.4	13.3	19.8	23.8	27.2	25.9	22.8	15.8	9.5	2.6	(32)	
	10%	WB	-1.0	-0.6	3.0	6.5	12.4	17.6	21.4	20.6	17.8	11.2	6.1	0.7	(33)	
		MCDB	-0.2	0.5	5.0	10.4	17.0	21.6	25.5	24.0	20.4	13.5	8.0	1.5	(34)	
(35) Mean Daily Temperature Range	MDBR		9.6	10.3	9.9	9.5	11.4	11.6	12.0	11.9	11.5	10.0	8.7	8.0	(35)	
	5% DB	MCDBR	9.0	11.3	13.1	17.2	18.6	17.1	14.9	14.9	15.2	15.3	13.0	8.5	(36)	
		MCWBR	7.6	8.6	8.8	10.1	10.2	9.5	8.0	8.2	8.9	9.5	8.7	7.0	(37)	
	5% WB	MCDBR	8.7	9.8	12.4	16.0	16.4	15.0	14.0	13.3	12.9	13.0	11.7	7.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.249	0.272	0.315	0.355	0.385	0.381	0.386	0.392	0.369	0.317	0.291	0.269	(40)	
	taud		2.412	2.361	2.357	2.364	2.331	2.371	2.380	2.385	2.420	2.552	2.514	2.419	(41)	
	Ebn at Noon		868	910	911	901	880	884	873	854	843	850	807	793	(42)	
	Edh at Noon		67	88	104	114	122	118	116	111	98	74	62	61	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.45	2.54	3.69	4.60	5.33	5.77	6.12	5.12	3.86	2.35	1.41	1.03	(44)	
	RadStd		0.16	0.23	0.34	0.46	0.40	0.40	0.25	0.37	0.26	0.30	0.16	0.10	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page